

ECE 5746 Lab 2

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1 RTL Design and Verification

Design. The multiplier (`mul.sv`, module `mult`) takes two 16-bit unsigned inputs `inp_a` and `inp_b` and registers their 32-bit product: on each rising edge of `clk`, `prod <= inp_a * inp_b`. An asynchronous, active-high `rst` clears `prod` to 0. The product is 32 bits wide, so the largest result ($0xFFFF \times 0xFFFF$) fits without overflow. Because the output is registered, synthesis times the path from the input ports through the multiplier logic to the `prod` flip-flops, which is the critical path reported in `syn_timing.rpt`.

Testbench. `tb_mul.sv` generates a 10 ns clock, holds `rst` high for the first 10 ns, and applies 5 input pairs, each held for 20 ns (2 clock cycles). The vectors cover small values, zero, the largest inputs (which exercise every partial-product bit and the top bits of `prod`), and a mixed value. The testbench waits one extra 20 ns after the last vector before `$finish`, so the last product is visible in the waveform. I checked each product in the QuestaSim waveform against the expected value in Table 1.

applied (ns)	inp_a	inp_b	expected prod	decimal	prod updates (ns)
0	0001	0002	00000002	2	15
20	0003	0004	0000000c	12	25
40	0000	0000	00000000	0	45
60	ffff	ffff	fffe0001	4,294,836,225	65
80	00c0	0acf	00081b40	531,264	85

Table 1: Test vectors from `tb_mul.sv`. Expected products are computed from the inputs; the waveform in Figure 1 shows the same values on `prod`.

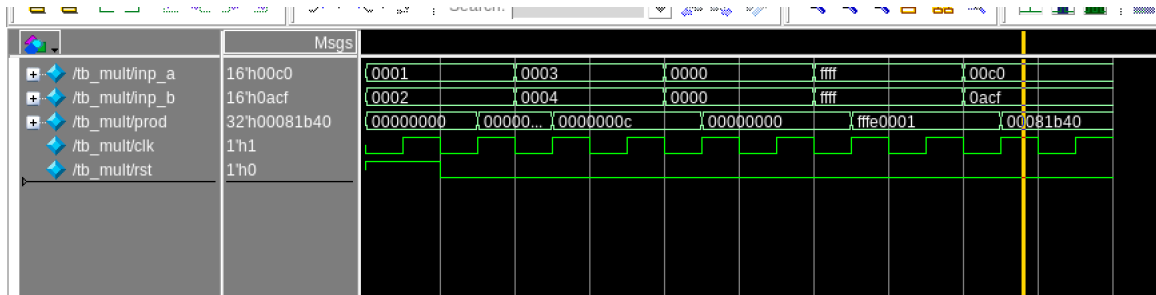


Figure 1: QuestaSim waveform of `tb_mult`: inputs, registered product, clock and reset.

Result. All 5 products match the expected values. While `rst` is high, `prod` stays at 0, so the first product appears at 15 ns, on the first rising edge after reset is released. After that, each new product appears on the first rising edge after the inputs change, which is the one-register latency of the design. $0xFFFF \times 0xFFFF = 0xFFFE0001$ confirms the full 32-bit result is kept.

2 SVT20: Fmax and Energy Sweep

For the standard- V_t cells (BWP20P90), I swept the clock period in `lab2.sh` and checked the worst slack in `syn_timing.rpt`. I use $clk_period_{min} = 0.33$ ns (chosen at 0.01 ns precision; a finer search showed 0.327 ns also meets timing), so $F_{max} = 1/0.33$ ns = 3030 MHz. I then synthesized the design at $1\times$ to $1000\times$ this clock period.

Energy per cycle is computed from `syn_power.rpt`: P_{Dyn} is switching plus internal power (mW), P_{Leak} is leakage power (reported in nW), and $E_{Dyn} = P_{Dyn} \times clk_period$, $E_{Leak} = P_{Leak} \times clk_period$, $E_{total} = E_{Dyn} + E_{Leak}$. With mW and ns, the energies are in pJ.

$\times$ min	clk (ns)	f (MHz)	slack (ns)	area (μm^2)	E_{Dyn} (pJ)	E_{Leak} (pJ)	E_{total} (pJ)
1.00	0.33	3030.3	0.00	525.7	0.9227	3.53e-04	0.9230
1.05	0.3465	2886.0	0.00	440.0	0.7540	2.69e-04	0.7543
1.10	0.363	2754.8	0.00	407.6	0.6951	2.45e-04	0.6954
1.15	0.3795	2635.0	0.00	380.5	0.6391	2.21e-04	0.6393
1.20	0.396	2525.3	0.00	371.8	0.6292	2.20e-04	0.6295
1.25	0.4125	2424.2	0.00	365.6	0.6150	2.20e-04	0.6153
1.30	0.429	2331.0	0.00	359.5	0.6045	2.26e-04	0.6047
1.40	0.462	2164.5	0.00	355.5	0.5983	2.41e-04	0.5985
1.50	0.495	2020.2	0.00	356.1	0.6014	2.59e-04	0.6017
1.75	0.5775	1731.6	0.00	352.1	0.5983	2.99e-04	0.5986
2.00	0.66	1515.2	0.00	348.2	0.5881	3.37e-04	0.5884
3.00	0.99	1010.1	0.00	327.4	0.5693	3.99e-04	0.5696
5.00	1.65	606.1	0.51	318.0	0.5231	6.07e-04	0.5237
10.00	3.3	303.0	2.16	318.2	0.5214	1.19e-03	0.5226
100.00	33	30.3	31.86	318.2	0.5214	1.19e-02	0.5333
1000.00	330	3.0	328.87	318.2	0.5214	1.19e-01	0.6404

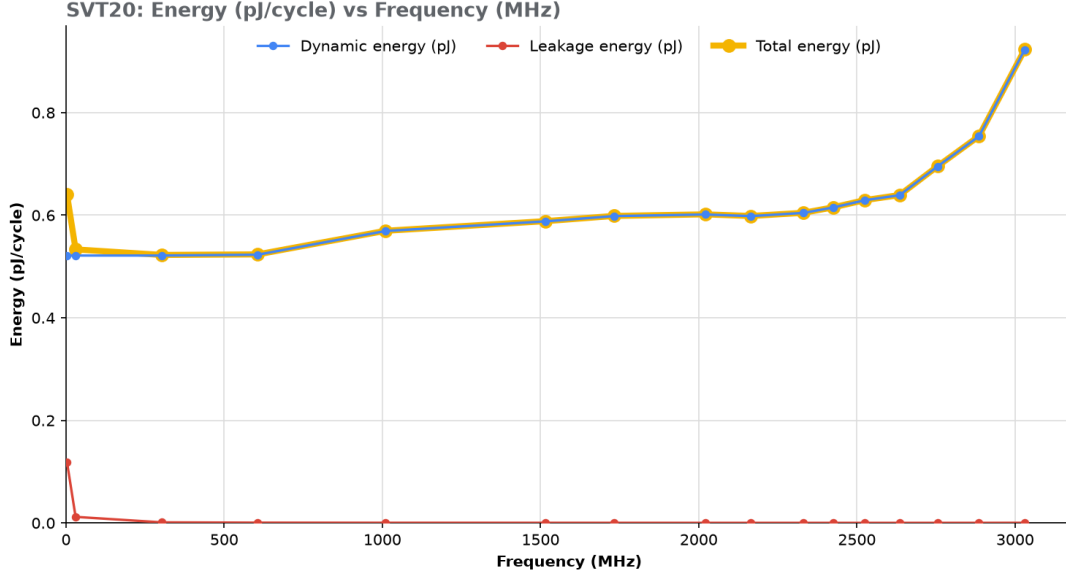


Figure 2: SVT20: dynamic, leakage and total energy per cycle vs frequency. Each marker is one synthesis run.

The minimum total energy is 0.5226 pJ/cycle at 303 MHz (3.3 ns, 10×). At F_{max} it is 0.9230 pJ/cycle.

3 LVT20: Fmax and Energy Sweep

For the low- V_t cells (BWP20P90LVT), I swept the clock period in `lab2.sh` and checked the worst slack in `syn_timing.rpt`. I use $clk_period_{min} = 0.244$ ns (0.244 ns meets timing and 0.242 ns does not), so $F_{max} = 1/0.244$ ns = 4098 MHz. I then synthesized the design at 1× to 1000× this clock period.

Energy per cycle is computed from `syn_power.rpt`: P_{Dyn} is switching plus internal power (mW), P_{Leak} is leakage power (reported in nW), and $E_{Dyn} = P_{Dyn} \times clk_period$, $E_{Leak} = P_{Leak} \times clk_period$, $E_{total} = E_{Dyn} + E_{Leak}$. With mW and ns, the energies are in pJ.

$\times$ min	clk (ns)	f (MHz)	slack (ns)	area (μm^2)	E_{Dyn} (pJ)	E_{Leak} (pJ)	E_{total} (pJ)
1.00	0.244	4098.4	0.00	525.2	0.9894	3.27e-03	0.9927
1.05	0.2562	3903.2	0.00	462.8	0.8306	2.77e-03	0.8334
1.10	0.2684	3725.8	0.00	412.7	0.7378	2.32e-03	0.7401
1.15	0.2806	3563.8	0.00	391.2	0.7158	2.17e-03	0.7180
1.20	0.2928	3415.3	0.00	377.7	0.6960	2.11e-03	0.6981
1.25	0.305	3278.7	0.00	367.7	0.6771	2.06e-03	0.6792
1.30	0.3172	3152.6	0.00	364.4	0.6690	2.09e-03	0.6711
1.40	0.3416	2927.4	0.00	354.8	0.6497	2.19e-03	0.6519
1.50	0.366	2732.2	0.00	355.5	0.6533	2.38e-03	0.6557
1.75	0.427	2341.9	0.00	352.6	0.6448	2.75e-03	0.6475
2.00	0.488	2049.2	0.00	346.9	0.6359	3.07e-03	0.6389
3.00	0.732	1366.1	0.00	320.1	0.6273	3.85e-03	0.6312
5.00	1.22	819.7	0.38	310.5	0.5649	5.78e-03	0.5706
10.00	2.44	409.8	1.61	310.7	0.5585	1.14e-02	0.5699
100.00	24.4	41.0	23.57	310.7	0.5583	1.14e-01	0.6720
1000.00	244	4.1	243.17	310.7	0.5583	1.14e+00	1.6953

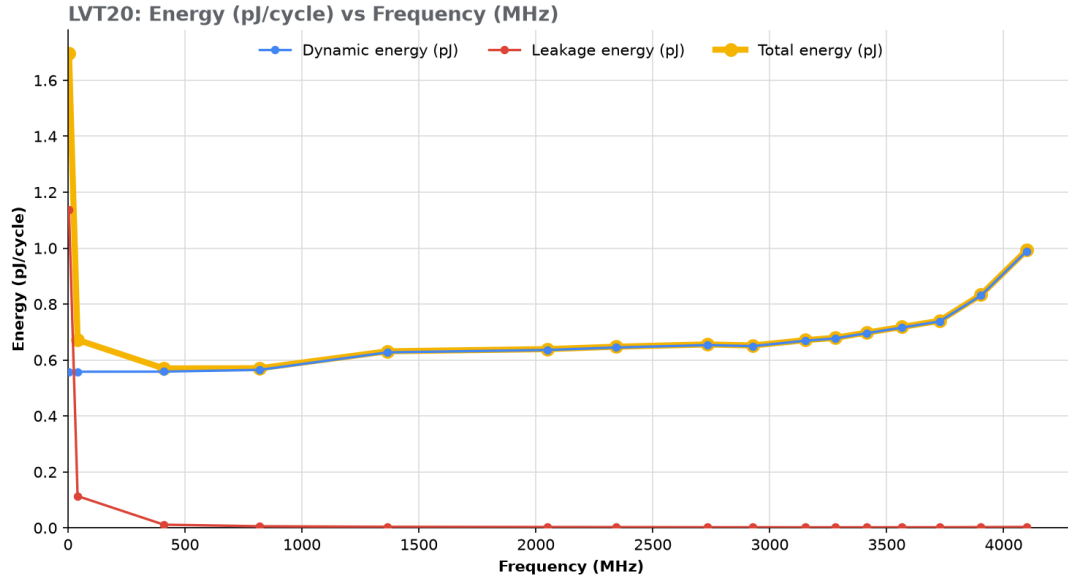


Figure 3: LVT20: dynamic, leakage and total energy per cycle vs frequency. Each marker is one synthesis run.

The minimum total energy is 0.5699 pJ/cycle at 410 MHz (2.44 ns, 10 $\times$). At F_{max} it is 0.9927 pJ/cycle.

4 LVT16: Fmax and Energy Sweep

For the low- V_t cells (BWP16P90LVT), I swept the clock period in `lab2.sh` and checked the worst slack in `syn_timing.rpt`. I use $clk_period_{min} = 0.22$ ns (chosen at 0.01 ns precision:

0.22 ns meets timing and 0.21 ns does not), so $F_{max} = 1/0.22 \text{ ns} = 4545 \text{ MHz}$. I then synthesized the design at $1\times$ to $1000\times$ this clock period.

Energy per cycle is computed from `syn_power.rpt`: P_{Dyn} is switching plus internal power (mW), P_{Leak} is leakage power (reported in nW), and $E_{Dyn} = P_{Dyn} \times clk_period$, $E_{Leak} = P_{Leak} \times clk_period$, $E_{total} = E_{Dyn} + E_{Leak}$. With mW and ns, the energies are in pJ.

$\times$ min	clk (ns)	f (MHz)	slack (ns)	area (μm^2)	E_{Dyn} (pJ)	E_{Leak} (pJ)	E_{total} (pJ)
1.00	0.22	4545.5	0.00	629.8	1.2096	9.50e-03	1.2191
1.05	0.231	4329.0	0.00	520.3	0.9739	7.21e-03	0.9811
1.10	0.242	4132.2	0.00	434.7	0.8027	5.30e-03	0.8080
1.15	0.253	3952.6	0.00	410.8	0.7529	5.03e-03	0.7580
1.20	0.264	3787.9	0.00	378.9	0.6766	4.44e-03	0.6811
1.25	0.275	3636.4	0.00	370.0	0.6639	4.32e-03	0.6682
1.30	0.286	3496.5	0.00	368.8	0.6607	4.55e-03	0.6652
1.40	0.308	3246.8	0.00	358.4	0.6443	4.62e-03	0.6490
1.50	0.33	3030.3	0.00	359.6	0.6491	5.15e-03	0.6543
1.75	0.385	2597.4	0.00	355.9	0.6464	5.89e-03	0.6523
2.00	0.44	2272.7	0.00	350.3	0.6499	6.60e-03	0.6565
3.00	0.66	1515.2	0.00	319.0	0.6178	7.85e-03	0.6256
5.00	1.1	909.1	0.36	310.7	0.5544	1.18e-02	0.5662
10.00	2.2	454.5	1.46	310.7	0.5544	2.35e-02	0.5779
100.00	22	45.5	21.26	310.7	0.5491	2.29e-01	0.7779
1000.00	220	4.5	219.26	310.7	0.5467	2.29e+00	2.8347

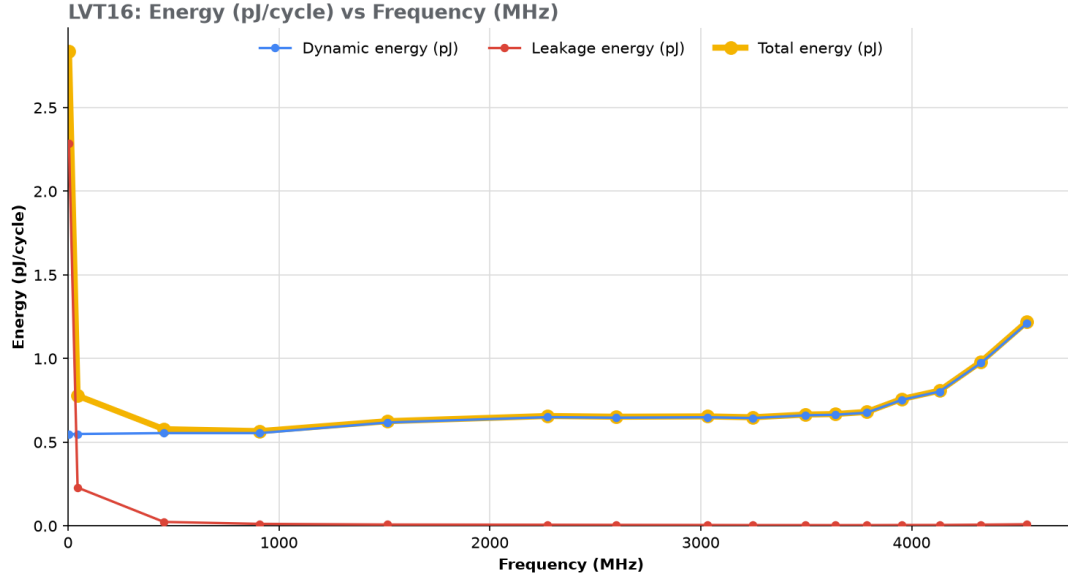


Figure 4: LVT16: dynamic, leakage and total energy per cycle vs frequency. Each marker is one synthesis run.

The minimum total energy is 0.5662 pJ/cycle at 909 MHz (1.1 ns, 5 $\times$). At F_{max} it is 1.2191 pJ/cycle.

5 PPA Trends and Optimal Operating Frequency

Below I discuss the performance, power and area trends for each graph, using `syn_area.rpt`, `syn_timing.rpt`, `syn_power.rpt` and the cell list in `syn.rpt`. Figure 5 summarizes how the netlist changes with frequency.

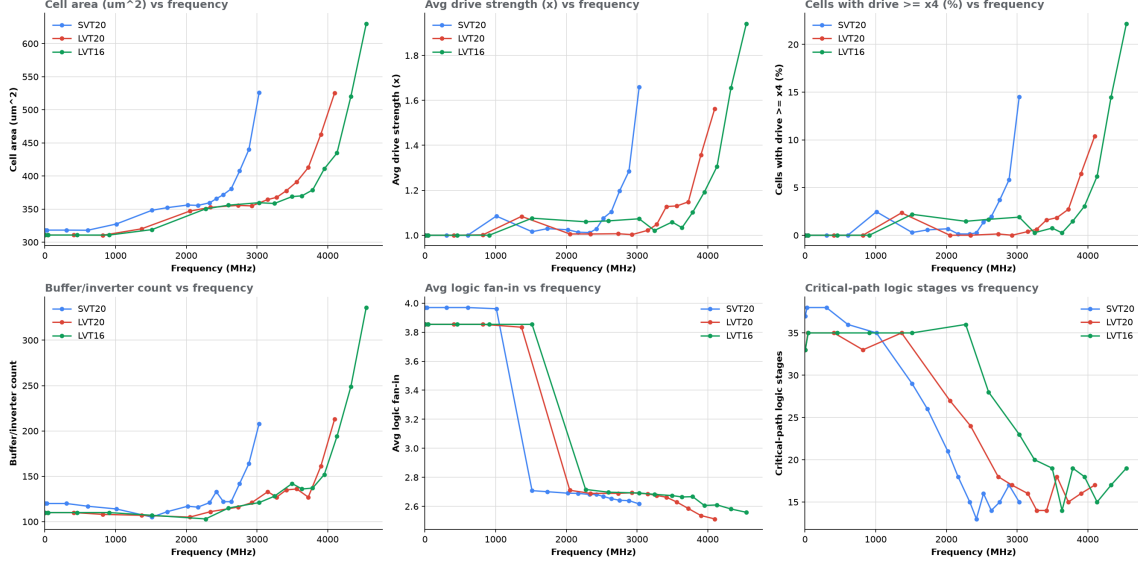


Figure 5: Area, drive strength, buffer/inverter count, fan-in and critical-path length vs frequency for all three libraries.

SVT20 (Figure 2)

Performance. At $F_{max} = 3030$ MHz the worst path starts at `inp_a[5]`, ends at `prod_reg[29]`, and passes through 15 cells with a data arrival time of 0.31 ns (`syn_timing.rpt`). The slack stays at 0 ns until 606 MHz ($5\times$), where it becomes 0.51 ns. Below that frequency the smallest netlist DC can build is already fast enough, so its critical path (37 cells, 1.11 ns arrival) no longer depends on the clock constraint.

Area. Total cell area rises from $318.2 \mu m^2$ at the slowest clock to $525.7 \mu m^2$ at F_{max} (+65%), and the cell count from 647 to 1148 (`syn_area.rpt`). Most of the growth is near F_{max} : area is $348.2 \mu m^2$ at $2\times$ and $365.6 \mu m^2$ at $1.25\times$. Non-combinational area stays at $36.5 \mu m^2$ (32 flip-flops for `prod`), so all of the extra area is combinational logic. Buffers and inverters grow from 120 to 208 (18.8 to $44.6 \mu m^2$), which DC adds to drive the larger loads on the critical paths.

Drive strength and fan-in. From the cell list in `syn.rpt`, at the slowest clock 100% of the combinational cells are D1 (average drive $1.00\times$). At F_{max} the average drive is $1.66\times$, 14.5% of cells are D4 or larger (29% of combinational area), and the largest cell is D12. The cells on the critical path average $2.6\times$ drive, compared with $1.0\times$ when relaxed. The gate mix also changes: the relaxed netlist uses 139 full/half-adder cells and high-fan-in complex gates (average fan-in 3.97, 84% with three or more inputs), while the F_{max} netlist uses only 28 adder cells, 292 XOR/XNR gates and more two- and three-input AOI/OAI gates

(average fan-in 2.62, 46% with three or more inputs). Gates with fewer inputs have fewer transistors in series and lower logical effort, and the restructured logic cuts the critical path from 37 to 15 cells. The switch happens between 1010 and 1515 MHz, where the average fan-in drops from 3.96 to 2.71 and the adder count from 139 to 119. This buys speed at the cost of more, larger cells.

Power and energy. Dynamic power is 2.796 mW at F_{max} and 0.0016 mW at 3.03 MHz, roughly proportional to frequency. Per cycle, however, E_{Dyn} rises from 0.521 pJ to 0.923 pJ (+77%), because the upsized, larger netlist switches more capacitance each cycle. Leakage power follows the cell area and size: 1070 nW at F_{max} and 361 nW when relaxed. Because E_{Leak} is leakage power times the clock period, it is only 0.04% of E_{total} at F_{max} but 18.6% at 3.03 MHz. The total energy curve therefore has a minimum of 0.5226 pJ at 303 MHz: faster than this, upsizing raises E_{Dyn} ; slower than this, leakage accumulates over the longer cycle.

f (MHz)	area	cells	buf/inv	avg drive	% D4+	max D	fan-in	FA/HA	XOR	path cells	arrival
3030	525.7	1148	208	1.66	14.5	12	2.62	28	292	15	0.31
2424	365.6	819	133	1.03	0.3	4	2.68	117	174	13	0.39
2020	356.1	770	117	1.02	0.7	4	2.69	117	174	21	0.48
1515	348.2	730	105	1.02	0.3	4	2.71	119	173	29	0.64
1010	327.4	642	114	1.09	2.5	4	3.96	139	1	35	0.97
606	318.0	644	117	1.00	0.0	1	3.97	139	1	36	1.12
303	318.2	647	120	1.00	0.0	1	3.97	139	1	38	1.12
3	318.2	647	120	1.00	0.0	1	3.97	139	1	37	1.11

Table 2: SVT20 netlist statistics from `syn_area.rpt`, `syn.rpt` and `syn_timing.rpt` (area in μm^2 , arrival in ns).

Critical-path cells at F_{max} (function $\times$ drive): XOR2x2 CKNx8 CKBx1 OAI22x1 FA1x1 XOR3x4 OAI21x2 IOA21x1 NR2x4 NR2x4 CKND2x4 INVx4 ND2x1 IA0I21x1 XOR2x1.

LVT20 (Figure 3)

Performance. At $F_{max} = 4098$ MHz the worst path starts at `inp_b[0]`, ends at `prod_reg[26]`, and passes through 17 cells with a data arrival time of 0.23 ns (`syn_timing.rpt`). The slack stays at 0 ns until 820 MHz ($5\times$), where it becomes 0.38 ns. Below that frequency the smallest netlist DC can build is already fast enough, so its critical path (33 cells, 0.82 ns arrival) no longer depends on the clock constraint.

Area. Total cell area rises from $310.7 \mu m^2$ at the slowest clock to $525.2 \mu m^2$ at F_{max} (+69%), and the cell count from 628 to 1254 (`syn_area.rpt`). Most of the growth is near F_{max} : area is $346.9 \mu m^2$ at $2\times$ and $367.7 \mu m^2$ at $1.25\times$. Non-combinational area stays at $36.5 \mu m^2$ (32 flip-flops for `prod`), so all of the extra area is combinational logic. Buffers and inverters grow from 110 to 213 (17.2 to $43.7 \mu m^2$), which DC adds to drive the larger loads on the critical paths.

Drive strength and fan-in. From the cell list in `syn.rpt`, at the slowest clock 100% of the combinational cells are D1 (average drive $1.00\times$). At F_{max} the average drive is $1.56\times$, 10.4% of cells are D4 or larger (20% of combinational area), and the largest cell is D12. The

cells on the critical path average $2.3\times$ drive, compared with $1.0\times$ when relaxed. The gate mix also changes: the relaxed netlist uses 139 full/half-adder cells and high-fan-in complex gates (average fan-in 3.85, 89% with three or more inputs), while the F_{max} netlist uses only 17 adder cells, 372 XOR/XNR gates and more two- and three-input AOI/OAI gates (average fan-in 2.51, 37% with three or more inputs). Gates with fewer inputs have fewer transistors in series and lower logical effort, and the restructured logic cuts the critical path from 33 to 17 cells. The switch happens between 1366 and 2049 MHz, where the average fan-in drops from 3.83 to 2.71 and the adder count from 139 to 118. This buys speed at the cost of more, larger cells.

Power and energy. Dynamic power is 4.055 mW at F_{max} and 0.0023 mW at 4.10 MHz, roughly proportional to frequency. Per cycle, however, E_{Dyn} rises from 0.558 pJ to 0.989 pJ (+77%), because the upsized, larger netlist switches more capacitance each cycle. Leakage power follows the cell area and size: 13400 nW at F_{max} and 4660 nW when relaxed. Because E_{Leak} is leakage power times the clock period, it is only 0.33% of E_{total} at F_{max} but 67.1% at 4.10 MHz. The total energy curve therefore has a minimum of 0.5699 pJ at 410 MHz: faster than this, upsizing raises E_{Dyn} ; slower than this, leakage accumulates over the longer cycle.

f (MHz)	area	cells	buf/inv	avg drive	% D4+	max D	fan-in	FA/HA	XOR	path cells	arrival
4098	525.2	1254	213	1.56	10.4	12	2.51	17	372	17	0.23
3279	367.7	820	127	1.05	0.6	8	2.67	116	176	14	0.29
2732	355.5	776	116	1.01	0.1	4	2.69	115	178	18	0.35
2049	346.9	732	105	1.01	0.0	2	2.71	118	174	27	0.47
1366	320.1	628	107	1.08	2.3	4	3.83	139	1	35	0.71
820	310.5	626	108	1.00	0.0	2	3.85	139	1	33	0.82
410	310.7	628	110	1.00	0.0	1	3.85	139	1	35	0.82
4	310.7	628	110	1.00	0.0	1	3.85	139	1	33	0.82

Table 3: LVT20 netlist statistics from `syn_area.rpt`, `syn.rpt` and `syn_timing.rpt` (area in μm^2 , arrival in ns).

Critical-path cells at F_{max} (function $\times$ drive): CKBx1 XNR2x1 OAI22x1 XOR2x1 XOR2x2 XNR2x2 XNR2x4 ND2x1 INVx1 AOI21x1 ND2x2 AOI21x4 OAI21x8 AOI21x4 OAI21x4 AOI21x1 XOR2x1.

LVT16 (Figure 4)

Performance. At $F_{max} = 4545$ MHz the worst path starts at `inp_a[13]`, ends at `prod_reg[27]`, and passes through 19 cells with a data arrival time of 0.20 ns (`syn_timing.rpt`). The slack stays at 0 ns until 909 MHz ($5\times$), where it becomes 0.36 ns. Below that frequency the smallest netlist DC can build is already fast enough, so its critical path (33 cells, 0.73 ns arrival) no longer depends on the clock constraint.

Area. Total cell area rises from $310.7 \mu m^2$ at the slowest clock to $629.8 \mu m^2$ at F_{max} (+103%), and the cell count from 628 to 1359 (`syn_area.rpt`). Most of the growth is near F_{max} : area is $350.3 \mu m^2$ at $2\times$ and $370.0 \mu m^2$ at $1.25\times$. Non-combinational area stays at $36.5 \mu m^2$ (32 flip-flops for `prod`), so all of the extra area is combinational logic. Buffers and

inverters grow from 110 to 336 (17.2 to $67.0 \mu m^2$), which DC adds to drive the larger loads on the critical paths.

Drive strength and fan-in. From the cell list in `syn.rpt`, at the slowest clock 100% of the combinational cells are D1 (average drive $1.00\times$). At F_{max} the average drive is $1.94\times$, 22.2% of cells are D4 or larger (45% of combinational area), and the largest cell is D16. The cells on the critical path average $3.7\times$ drive, compared with $1.0\times$ when relaxed. The gate mix also changes: the relaxed netlist uses 139 full/half-adder cells and high-fan-in complex gates (average fan-in 3.85, 89% with three or more inputs), while the F_{max} netlist uses only 18 adder cells, 302 XOR/XNR gates and more two- and three-input AOI/OAI gates (average fan-in 2.56, 38% with three or more inputs). Gates with fewer inputs have fewer transistors in series and lower logical effort, and the restructured logic cuts the critical path from 33 to 19 cells. The switch happens between 1515 and 2273 MHz, where the average fan-in drops from 3.85 to 2.72 and the adder count from 139 to 121. This buys speed at the cost of more, larger cells.

Power and energy. Dynamic power is 5.498 mW at F_{max} and 0.0025 mW at 4.55 MHz, roughly proportional to frequency. Per cycle, however, E_{Dyn} rises from 0.547 pJ to 1.210 pJ (+121%), because the upsized, larger netlist switches more capacitance each cycle. Leakage power follows the cell area and size: 43200 nW at F_{max} and 10400 nW when relaxed. Because E_{Leak} is leakage power times the clock period, it is only 0.78% of E_{total} at F_{max} but 80.7% at 4.55 MHz. The total energy curve therefore has a minimum of 0.5662 pJ at 909 MHz: faster than this, upsizing raises E_{Dyn} ; slower than this, leakage accumulates over the longer cycle.

f (MHz)	area	cells	buf/inv	avg drive	% D4+	max D	fan-in	FA/HA	XOR	path cells	arrival
4545	629.8	1359	336	1.94	22.2	16	2.56	18	302	19	0.20
3636	370.0	839	136	1.03	0.2	4	2.66	113	180	14	0.26
3030	359.6	770	121	1.07	1.9	4	2.69	117	174	23	0.31
2273	350.3	717	103	1.06	1.5	4	2.72	121	170	36	0.42
1515	319.0	625	107	1.08	2.2	4	3.85	139	1	35	0.64
909	310.7	628	110	1.00	0.0	1	3.85	139	1	35	0.73
455	310.7	628	110	1.00	0.0	1	3.85	139	1	35	0.73
5	310.7	628	110	1.00	0.0	1	3.85	139	1	33	0.73

Table 4: LVT16 netlist statistics from `syn_area.rpt`, `syn.rpt` and `syn_timing.rpt` (area in μm^2 , arrival in ns).

Critical-path cells at F_{max} (function $\times$ drive): CKNx2 INVx4 BUFFx4 XNR2x1 NR2x2 NR2x4 XNR2x4 XNR2x8 XOR2x8 XOR2x4 XNR2x4 ND2x2 INVx1 AOI21x4 OAI21x4 AOI21x8 OAI21x4 AOI21x1 XNR2x1.

Optimal operating frequency

Library	(a) Area-constrained	(b) Timing-critical	(c) Energy-constrained
SVT20	606 MHz ($318.0 \mu m^2$)	3030 MHz ($525.7 \mu m^2$)	303 MHz (0.5226 pJ)
LVT20	820 MHz ($310.5 \mu m^2$)	4098 MHz ($525.2 \mu m^2$)	410 MHz (0.5699 pJ)
LVT16	909 MHz ($310.7 \mu m^2$)	4545 MHz ($629.8 \mu m^2$)	909 MHz (0.5662 pJ)

SVT20. (a) For an area-constrained design I would run at 606 MHz (1.65 ns). This is the fastest point whose area is within 1% of the minimum ($318.0 \mu m^2$); a slower clock gives no further area savings, only lower throughput. (b) For a timing-critical design I would run at $F_{max} = 3030$ MHz, which costs 65% more area and 77% more energy per cycle than the minimum. (c) For an energy-constrained design I would run at 303 MHz (3.3 ns), the minimum of E_{total} (0.5226 pJ/cycle); since 606 MHz is within 2% of this energy (0.5237 pJ), it is also a good choice with higher throughput.

LVT20. (a) For an area-constrained design I would run at 820 MHz (1.22 ns). This is the fastest point whose area is within 1% of the minimum ($310.5 \mu m^2$); a slower clock gives no further area savings, only lower throughput. (b) For a timing-critical design I would run at $F_{max} = 4098$ MHz, which costs 69% more area and 74% more energy per cycle than the minimum. (c) For an energy-constrained design I would run at 410 MHz (2.44 ns), the minimum of E_{total} (0.5699 pJ/cycle); since 820 MHz is within 2% of this energy (0.5706 pJ), it is also a good choice with higher throughput.

LVT16. (a) For an area-constrained design I would run at 909 MHz (1.1 ns). This is the fastest point whose area is within 1% of the minimum ($310.7 \mu m^2$); a slower clock gives no further area savings, only lower throughput. (b) For a timing-critical design I would run at $F_{max} = 4545$ MHz, which costs 103% more area and 115% more energy per cycle than the minimum. (c) For an energy-constrained design I would run at 909 MHz (1.1 ns), the minimum of E_{total} (0.5662 pJ/cycle).

Across libraries. LVT16 reaches the highest F_{max} (4545 MHz), so it is the choice for a timing-critical design. SVT20 has the lowest minimum energy (0.5226 pJ/cycle) because its leakage is much lower (SVT20: 1070 nW, LVT20: 13400 nW, LVT16: 43200 nW at F_{max}), so it is the choice for an energy-constrained design. The minimum areas are close (SVT20: 318.0, LVT20: 310.5, LVT16: $310.7 \mu m^2$), with LVT20 the smallest. At the slowest clock, where the LVT16 and LVT20 netlists have almost the same area (310.7 and $310.7 \mu m^2$), LVT16 leaks 10400 nW versus 4660 nW, so the BWP16 cells leak $2.2\times$ more than the BWP20 cells of the same V_t in exchange for their higher speed. Low- V_t cells switch faster but leak more, which is why the LVT libraries have higher F_{max} and a steeper rise in total energy at low frequency.